

News Frame Analysis: An Inductive Mixed-method Computational Approach

Dror Walter & Yotam Ophir

To cite this article: Dror Walter & Yotam Ophir (2019) News Frame Analysis: An Inductive Mixed-method Computational Approach, *Communication Methods and Measures*, 13:4, 248-266, DOI: [10.1080/19312458.2019.1639145](https://doi.org/10.1080/19312458.2019.1639145)

To link to this article: <https://doi.org/10.1080/19312458.2019.1639145>

 [View supplementary material](#) 

 Published online: 23 Jul 2019.

 [Submit your article to this journal](#) 

 Article views: 8827

 [View related articles](#) 

 [View Crossmark data](#) 

 Citing articles: 93 [View citing articles](#) 



News Frame Analysis: An Inductive Mixed-method Computational Approach

Dror Walter ^a and Yotam Ophir ^b

^aCommunication, Georgia State University, Atlanta, USA; ^bCommunication, University at Buffalo, State University of New York, Buffalo, USA

ABSTRACT

Framing is one of the most central, applicable, and contested theories in communication research. At the heart of the debate on framing is the question of operationalizing and measuring emphasis frames. We harness novel computational tools to propose a new method for inductive identification of frames. We argue and demonstrate that frame elements could be identified using topic modeling, and that frame elements can then be automatically grouped into frame “packages” using community detection techniques applied to the topic network. Building upon recent conceptual and methodological developments in framing research, we introduce a new approach, the Analysis of Topic Model Networks (ANTMN). We demonstrate the applicability of our method in case studies where framing theory is developed and fairly consistent, and in exploratory ones where it is not, using three diverse U.S. news corpora: the coverage of political candidates in Senate races ($n = 8,337$ articles), foreign nations ($n = 18,216$), and infectious diseases and epidemics ($n = 5,005$). We conclude by discussing the theoretical, methodological, and practical implications of ANTMN.

The exponential growth in social data in recent years and the development of novel computational tools for the analysis of big data allowed researchers to revisit looming questions about prominent communication theories, supplementing, challenging, or augmenting traditional methods (van Atteveldt & Peng, 2018). One of the most prominent, yet contested, traditional theories in communication research is framing (Matthes, 2009). Questions of how to conceptualize, identify, and measure the prevalence of frames have been a topic of heated debates for decades (Cacciatore, Scheufele, & Iyengar, 2016; D’Angelo, 2018; van Gorp, 2010). In this study, we harness computational techniques for big data analysis, topic modeling and network analysis, to examine whether frames could be identified in an inductive, unsupervised, and valid manner from large corpora.

We argue that recently suggested computer-assisted approaches, such as the treatment of individual topics as frames (DiMaggio, Nag, & Blei, 2013; Klebanov, Diermeier, & Beigman, 2008), could be inconsistent and unreliable due to researchers’ influence at different stages of modeling, and the tendency of some topics to be narrow and specific (depending, in part, on model parameters; Maier et al., 2018). Instead, following prior definitions of frames as consisting of recurring linguistic elements or “devices” that cluster over time and articles into frame “packages” (Baden, 2018; D’Angelo, 2018; van Gorp, 2010), we suggest that news frames could be operationalized as communities in a network of topics. We argue that our approach removes some of the bias resulting from treating topics as frames, while remaining conceptually consistent with the broad nature of news frames and with recent

CONTACT Dror Walter  dwalter2@gsu.edu  25 Park Place, Atlanta, GA 30303, United States

Sample code for topic model network creation (ANTMN) can be accessed in <https://github.com/DrorWalt/ANTMN>

Color versions of one or more of the figures in the article can be found online at www.tandfonline.com/hcms.

 Supplemental data for this article can be accessed [here](#).

© 2019 Taylor & Francis Group, LLC

conceptual and methodological developments, treating frame as matrices of frame packages and frame devices (van Gorp, 2010). In addition, our approach allows researchers to benefit from the vast potential of the rapidly evolving field of network analysis techniques to enrich their analysis of frames. We call our approach the Analysis of Topic Model Networks (ANTMN).

Following our discussion of theoretical and methodological reasoning behind ANTMN, we demonstrate the method's ability to identify frames, and measure their prevalence using three diverse large corpora, spanning the news coverage of political candidates (Walter, 2018), foreign countries (Walter, Sheaffer, Nir & Shenhav, 2016), and infectious diseases (Ophir, 2018a). We demonstrate the potential of ANTMN to bolster framing research by examining our inductive results in light of previous studies post-modeling. We do so in case studies where research has already pointed to fairly-consistent patterns (the use of strategy and issue frames during political campaigns; Cappella & Jamieson, 1997), where research on framing is still emerging and in need of validation (the use of soft and hard power perspectives in news coverage of foreign countries; Anholt, 2009; Hayden, 2012; Kaneva, 2011), and where research is relatively scarce and more exploratory work is needed (the news coverage of epidemic; *The Authors*, 2018b). We conclude by discussing the advantages and disadvantages of ANTMN in comparison with other available techniques.

Framing Theory: Conceptual and Methodological Debates and Challenges

At the heart of framing theory is the notion that the selective presentation of information can influence attitudes, beliefs, decisions and behaviors (Chong & Druckman, 2007). In contrast to agenda-setting theory (McCombs & Shaw, 1972) and its oft-cited emphasis on the media telling people “what to think about” (Cohen, 1965), framing is concerned with the media’s ability to tell audiences “how to think” about an issue, by presenting selective and limited accounts of issues and events (Cacciatore et al., 2016). While the amount of work dedicated to framing in different fields, including psychology, sociology, economics, and communication, had demonstrated its applicability and flexibility, it also led the theory to become highly contested and controversial – a “fractured paradigm” (Entman, 1993).

Some of the conceptual dispute over framing surround two competing approaches, “equivalency framing” and “emphasis framing”. Equivalency framing involves the presentation of different, yet logically equivalent, information to produce framing effects (Cacciatore et al., 2016). A paragon of equivalency framing is Prospect Theory and its demonstration of the effect of gain and loss frames (Kahneman & Tversky, 1984). The precision required by this approach to framing makes it applicable mostly to controlled environments such as tailored communication (e.g., Curtis, 2012) where message designers can meticulously alter and examine the effect of intentional word choices on audiences. Such practice, while in use by journalists in some cases, does not often result in news articles that are equivalent on all but the framing device. In other words, news articles often differ on more than just isolated framing elements.

Conceptualizing Emphasis Frames

Since “the events and issues that are the subject of most news stories do not lend themselves to setting up mathematically equivalent frames” (Shah, McLeod, Gotlieb, & Lee, 2009, p. 87), a more suitable approach for understanding news frames is through emphasis framing (D’Angelo, 2018). Emphasis framing is based on the premise that the selective presentation of events and issues through a focus on different arguments, perspectives, and facts could influence audiences’ thinking and reactions (Entman, 1993; Gitlin, 1980). Scholars and journalists alike have argued that framing choices are an essential part of journalistic routines (van Gorp, 2010), an “unavoidable reality of the public communication process” (Nisbet, 2010, p. 44), that is guided, in part, by ideology, news-worthiness, and other factors (Galtung & Ruge, 1965; Gitlin, 1980).

Research on emphasis framing often distinguishes between “generic” news frames and “topic-specific” news frames (D’Angelo, 2018). To avoid confusions, we refer to the latter as “context-specific” to avoid confusion with the term “topic” as it’s used in topic modeling. Generic news frames are generalizable across

events and contexts. They can be seen as established routines used by journalists to cover new events (Tuchman, 1973), and could therefore be understood as “what journalists do” in order to frame issues (D’Angelo, 2018, p. xxxi). These include, for example, Iyengar’s episodic vs. thematic news frames (Iyengar, 1996), or the frames of conflict, human interest, economic consequences, and morality (Semetko & Valkenburg, 2000). Context-specific news frames (De Vreese, 2005), on the other hand, are more pertinent to specific contexts than others. These include, for example, Nelson and colleagues’ analysis of media coverage of social protest (Nelson, Clawson, & Oxley, 1997) or De Vreese’s analysis of news coverage of the European Union’s economy (De Vreese, 2010). However, as the application of these frames eventually depends on journalists’ decision making, context-specific news frames may be adapted to other contexts as well. As argued by D’Angelo (2018, p. xxxii), “at bottom, the analytical difference between context-specific and generic news frames is more subtle than stark”. Therefore, in this study, we will look broadly at emphasis framing, without limiting our approach to one category or another. In the following parts we focus on recent methodological developments and debates surrounding the extraction of frames from texts.

Identifying and Measuring Emphasis Frames

In framing research, a deductive strategy is commonly used, where researchers identify predefined, theoretically-based, and limited sets of frames in corpora (Matthes & Kohring, 2008). However, whether researchers apply generic or context-specific news frames on new sets of documents, the adaptation of prior knowledge limits the lenses through which coverage could be analyzed, in a way that is often biased by the background, knowledge, and expectations of the researchers (van Gorp, 2010).

One way to diminish bias is to focus on identifying frame elements or devices instead of looking for the full frame package (Gamson & Modigliani, 1989), that is, to code not for the existence of a full frame in a text, but rather of manifested frame elements. These elements include, but are not limited to, word choices, metaphors, catchphrases, exemplars, descriptions, arguments, and visual icons (Baden, 2018; van Gorp, 2007). For example, Cappella and Jamieson (1997) identified strategy framing by looking for linguistic devices such as “horse race” or games and war metaphors. However, while this approach reduces some bias by inviting researchers to rely on quantifiable textual and visual “evidence”, it still relies on the assumptions that the frames chosen a priori by the researcher suit the context under investigation and that frames currently available in the literature are adequate to describe all contexts (Matthes & Kohring, 2008). Moreover, such an approach may result in researchers tending to find the frames they were consciously or unconsciously looking for (Tankard, 2001).

These challenges led scholars to look for ways to develop inductive methods for identifying frame elements. For example, van Gorp (2010) suggested conducting open-coding of texts without relying on predefined coding instruments, identifying various frame devices in the texts under consideration, and then arranging the devices found around “overarching ideas” that could be analyzed as complete frame packages (p. 95). Based on Gamson & Lasch’s approach (cited in Gamson & Modigliani, 1989) he suggested arranging the devices in a framing matrix, where each row represents a frame package and each column a list of framing devices by which the frame manifests itself in texts. While a step in the right direction, this approach still relies heavily on pre-existing knowledge and theory, thus presenting subjectivity to the identification and clustering of frame elements around overarching ideas.

As opposed to Gorp’s inductive-then-deductive approach, a purely inductive approach does not assume devices are connected based on theory but based on their inter-relationships within a given corpus. Such approach assumes that frame elements that tend to systematically appear together (co-occur) across multiple texts are thematically connected (Matthes & Kohring, 2008) and that frame elements that share similar vocabularies are connected, consistent with the definition of frames as news patterns that “repeatedly invoke the same objects and traits, using identical or synonymous

words and symbols in a series of similar communications that are concentrated in time” (Entman, Matthes, & Pellicano, 2009, p. 177).

Several approaches were taken toward conducting inductive framing analysis. Some researchers conducted semi-inductive analysis, first manually coding frame elements based on theory and then clustering the elements together automatically to identify their manifestation into frame packages (Boydston & Glazier, 2013). These researchers have used clustering techniques such as principal-components and varimax rotation (Semetko & Valkenburg, 2000), hierarchical clustering (Matthes & Kohring, 2008), and multiple dimensional scaling (van Gorp, 2005). However, the reliance on manual coding of frame elements meant that these scholars still depended on prior knowledge. In addition, these procedures require vast resources in terms of time and money.

Unsupervised Learning Approaches to Frame Analysis

More recently, scholars have suggested using unsupervised machine learning approaches that substantially reduce the reliance on manual coding. This was done mainly using two approaches; semantic network analysis and topic modeling. We draw on the first approach for the graph-representation of frame elements, and the document-based co-occurrence conceptualization. We draw on the second approach for the identification of frame elements which we model in the network.

Semantic Network Analysis. The first approach used to inductively identify frames was network analysis, mostly using semantic and lexical network analyses (Baden, 2018; Diesner & Carley, 2011; Doerfel & Barnett, 1999; van Atteveldt, 2008). A special case of generalized mathematical networks, these networks use semantic units (such as words or concepts) as nodes and the relationship between them (such as co-occurrence within documents or smaller text units defined by the researchers) as edges. Mapping texts into networks turns large corpora into manageable graphic representations. While not all the studies done with these tools examined the results in light of framing theory, some researchers have argued that clustering techniques could identify groups of words in a network that tend to appear together, and that these groups manifest frames (Baden, 2018).

Network based approaches have some clear advantages, on which we elaborate more in the discussion section. The main advantage is that the use of network analysis provides a highly efficient graphic representation for otherwise complex data, as well as a potential to borrow indicators from the rapidly evolving network theory in areas such as neurological and social network analyses, for understanding the relationships between words (see also Rubinov & Sporns, 2010). However, this approach has several disadvantages. First, networks that use words as nodes may be limited due to a reliance on “low-level linguistic results that are difficult to relate to theoretical concepts” (van Atteveldt, 2008, p. 67). This can also complicate additional analyses following the identification of frames in the corpus, such as comparative analysis of frames and statistical testing of hypotheses. Alternatively, networks that use pre-defined concepts as nodes (instead of individual words) are limited due to reliance on dictionary methods, that require pre-existing area-specific knowledge, or on a qualitative preliminary analysis in search of keywords that correspond to pre-defined frames (see Associative framing; van Atteveldt, 2008). Second, at the moment, techniques that are publicly-available in popular coding languages like R and Python are mostly limited to single-membership communities, which means that each word could be attributed to only one community (Yang, Algesheimer, & Tessone, 2016). However, this limitation might be temporary, as alternative approaches for multiple-membership are being developed (Yang et al., 2016).

In currently available multiple-membership methods, such as topic modeling (discussed next), each word has a probability to appear in each topic, thus allowing words to have different meanings in different contexts (Blei, 2012). The lack of multiple-membership in many current networks implementations could threaten the validity of network analysis when assigning words into clusters in cases of polysemy (e.g., financial “bank” vs. river “bank”) or when a word plays a different role as a linguistic device in different frames (e.g., “unequal” education system in issue frames vs. “unequal”

television air time in strategy frames). Finally, the use of semantic networks in social science research is currently limited. Topic modeling that has emerged as a leading unsupervised machine learning technique among computational communication scholars (Maier et al., 2018). The prominence of topic modeling results in rapid developments of new and more advanced techniques such as Structural Topic Modeling (Roberts et al., 2014), while the parallel development in semantic networks is currently slower. We therefore discuss below the advantages and disadvantages of topic modeling for frame analysis.

Topic Modeling. Topic modeling is an unsupervised method that uses a Bayesian generative approach for text analysis, in which the model mimics the writing process of a given corpus of documents (Blei, Ng, & Jordan, 2003). “Topics” are defined as sets of frequency distributions of words, based on co-occurrences (Blei, 2012). Further elaboration on the conceptualization and operation of topic modeling can be found in the method section. DiMaggio et al. (2013) have argued that topic modeling could be used for automatically identifying frames in texts, praising the method’s inductive (reducing researcher bias) and automated procedure (reducing costs and time). In addition, topic modeling allows for linguistic flexibility (as words can have different meanings in different contexts) and relative transparency, compared to most clustering methods that do not make the clustering process and choices apparent to users). However, the use of topic modeling for framing analysis raises some challenging questions (Maier et al., 2018).

DiMaggio and colleagues have argued that “many topics may be viewed as frames” (DiMaggio et al., 2013, p. 578), despite the fact that the theoretical meaning of topics remains largely unclear (Maier et al., 2018). We find several reasons to believe topics could not be conceptualized as news frames. First, the content and scope of topics may change based on the resolution chosen by the researcher (i.e., the number of topics). Second, topics in the same model could be either general or context-specific. Third, it is often the case that some topics in a model represent mere linguistic artifacts that are unintelligible for human readers (Mimno & Blei, 2011). To sum, it is unclear whether topics should be interpreted as frame devices, frame packages, or some other construct. One solution would be to connect topics into larger themes or meta-topics. This can be done manually by examining the topics, though the disadvantages are similar to all manual clustering of frame elements mentioned earlier (including researcher bias, efficiency, reproducibility, and the challenges of identifying themes and co-occurrence in large corpora). Such clustering can be also done automatically. For example, Puschmann and Scheffler (2016) used hierarchical cluster dendrogram to plot the similarity between different topics. However, such approaches prevent researchers from using the rich and sophisticated statistical analyses that are provided in network analysis, and semantic network analysis specifically.

The ANTMN Three-step Approach to Frame Analysis

Keeping the limitations of each approach in mind, and more importantly, the advantages of each, we suggest an alternative inductive approach aimed at combining topic modeling and network analysis for the purpose of frame analysis. Our approach consists of three steps. First, we identify frame elements using topic modeling, similarly to what was done by DiMaggio et al. (2013), but without equating individual topics to frames. Instead, we remain agnostic as to whether topics found consist of frame elements in a limited context, a full frame package, or serve as a combination of the two. Second, similar to the semantic network procedure (Baden, 2018; van Atteveldt, 2008), we situate the topics in a network structure, in which the topics serve as nodes and the relationships between them (edges) are calculated by their co-occurrence over documents. Third, and based on previous conceptualizations of frame packages (van Gorp, 2010), we use various community detection techniques to cluster together these topics into coherent frames (i.e., framing packages).

It is important to note that due to the sensitivity of computational models to corpus complexity and to the chosen resolution (e.g., number of topics), the theoretical meaning of individual topics may vary. In some cases, topics may represent a single frame element specific to a single context (e.g., the poll results for a specific political candidate), a single frame elements in multiple contexts

(the poll results for several candidates), multiple frame elements in a single context (poll results and fundraising for specific candidate), or multiple elements of multiple contexts (poll results and fundraising for several candidates).

We argue that this three-step procedure is congruent with the definition of framing by scholars working within the emphasis framing approach, which conceptualized framing as the constant use of language elements across texts and time (Entman et al., 2009). To examine our method and its applicability, we employ ANTMN independently on three corpora from diverse news media case studies: the coverage of political candidates during elections, the coverage of epidemics, and the coverage of foreign countries.

Method

Case Studies

In order to examine ANTMN performance we gathered data from three corpora relating to three case studies in different contexts. To better understand the method's performance and applicability in various communication research circumstances; we chose one case study where research has already pointed to fairly consistent patterns (the use of strategy and issue frames during political campaigns; Cappella & Jamieson, 1997), one where research on framing is still emerging (the use of soft and hard power perspectives in news coverage of foreign countries; Anholt, 2009; Hayden, 2012; Kaneva, 2011), and one where research is relatively scarce and more exploratory work is needed (the news coverage of epidemic; *The Authors*, 2018b). We emphasize that our method is exploratory and inductive, and therefore, while prior empirical findings and theoretical conceptualizations could aid the interpretation of the results, we do not use prior knowledge to set predictions or hypotheses in advance. Therefore, the purpose of the following sections is not to form a solid ground for a set of hypotheses, but instead is aimed at allowing the readers a better understanding of the state of knowledge around these topics, and, as a result, the ways in which they fit our methodological needs in this manuscript, and the potential contribution of the suggested method in these contexts.

The case study of coverage of **political candidates** during electoral campaigns was chosen as our example for a topic that was thoroughly studied in the past, and where researchers share a general agreement regarding the expected frame elements and packages used by the news media, namely issue and strategy frames (Cappella & Jamieson, 1997). Issue frames focus on decision making and argumentation related to policy. Strategy frames focus on winning, losing, self-interest, and the characteristics of candidates (Cappella & Jamieson, 1997). The importance of framing in this context is exemplified in empirical studies that consistently show strategy framing to increase political cynicism, reduce substance-based political knowledge, and discourage positive evaluations of news items (Zoizner, 2018).

We chose a corpus containing all news coverage of three 2016 Senate candidates gathered from the LexisNexis database. For each candidate, a search was performed using the candidate's full name, with a time-frame of six months prior to the elections. Search was performed for all U.S. Newspapers and Wire Services (for the full list of outlets see Lexis-Nexis code: 140,954). Cumulatively, we identified mentions of the three candidates in 570 news sources, including local news, national newspapers, wire services and government publications. To choose the candidates, we selected the most covered Senate candidates during the latest election cycle. To keep the corpus size relatively small and thematically consistent, we omitted candidates who competed for the presidential nomination, such as Marco Rubio. The three Senate candidates that were chosen for the model were Kelly Ayotte (NH; 2,622 articles), Ron Johnson (WI; 2,932 articles), and Rob Portman (OH; 2,783 articles).

For the second case study we chose a context where researchers have pointed to news media coverage patterns, but where it is not yet known whether these patterns constitute news frames, the coverage of **foreign countries** in American media. A prominent research approach to this question pointed to the journalistic focus on **soft and hard power** (Hayden, 2012; Nye, 2004). While other

countries' hard power is portrayed by the US media through stories on military force and coercion, stories about soft power describe other countries' culture, art, music, performance in sports, and their economy (Nye, 2011). Despite being a major field of scientific inquiry due to its expected effects on public opinion about other countries and about foreign affairs policy, especially in the realm of news media (Walter, Sheaffer, Nir, & Shenhav, 2016), very little is known about whether or not the focus on soft and hard power could be seen as news frames (Ji, 2017).

For this case study, we chose a corpus retrieved from similar LexisNexis sources containing the coverage of three countries between the years 2010–2015. Due to the volume of coverage we limited the current analysis to the prominent newspaper The Washington Post – retrieving both its online and offline coverage. Countries' coverage was retrieved by searching for the country name and its demonym (the name collectively given to its people in English). To test our method, we chose three countries from three different continents, that were included in Inglehart's Word Value Survey (Inglehart & Welzel, 2010): Egypt (Africa; 7,728 articles), Brazil (South America; 5,068 articles) and Switzerland (Europe; 5,420 articles).

The last case study involved the news coverage of **epidemics**. Despite the media's central role in communicating risks to the public during outbreaks of infectious diseases (Reynolds & Seeger, 2014), little systematic understanding of journalistic practices exists (The authors, 2018a). While researchers did attempt to employ theories of framing to the study of news coverage of epidemics, many approaches to frames, some content-specific and some generic, were used inconsistently (e.g., Yan An & Gower, 2009; Shih, Wijaya, & Brossard, 2008; Tian & Stewart, 2005; Wu, 2006), leading to an inability to form a coherent understanding of the news media during epidemics (The authors, 2018a). To examine ANTMN in such situation, where little is known about journalistic use of frames, we chose a corpus consisting of 5,005 articles (1,798 on H1N1, 2,122 on Ebola, and 1,095 on Zika). For each disease, all articles published in four leading newspapers (New York Time, Washington Post, USA Today, and Wall Street Journal) were collected from LexisNexis. The diseases were chosen to represent three large scale epidemics in the 21st century that exhibited cases in the United States. The H1N1 corpus spanned between January 9, 2009, and September 8, 2010, a month after WHO declared the epidemic was over. The Ebola corpus spanned between March 20, 2014, and July 16, 2015, a month after WHO declared Liberia to be Ebola-free. The Zika corpus spanned between December 29, 2015, and December 7, 2016, a month after WHO declared that Zika was no longer a global threat.

Procedures and Measures

Topic Modeling

Topic modeling is a semi-automated, unsupervised method for the analysis of textual content (Blei, 2012). Topic models use a Bayesian generative approach, in which the model attempts to “mimic” the writing process of a given corpus of documents (Blei et al., 2003). In other words, the algorithm extracts a set of topics from which the given corpus could have been created. “Topics” are statistical entities, sets of frequency distributions of words, based on the linguistic assumption of co-occurrence, that is, that words that are being used more frequently in the same documents also associate thematically. In topic models, every word in the corpus has a probability of appearing in each of the topics, and every document is composed as a mixture of all topics (Grimmer & Stewart, 2013). Topic modeling is a “bag-of-words” approach, which means that narrative, location in the text, and syntax are not taken into consideration (Wallach, 2006).

In this work, we use Latent Dirichlet Allocation (LDA) and Gibbs Sampling (LDA, see Blei et al., 2003), using the LDA and LDATuning packages for R.¹ First, texts were pre-processed by removing stop-words, converting capital letters to lowercase, removing punctuations and numbers, and removing words that appeared in more than 99% of the documents, or in less than .5% (in this order), based on guidelines suggested by Maier et al. (2018). We refrained from lemmatization or stemming due to its possible negative influence on topic stability (Schofield, Thompson, & Mimno,

2017). We estimated separate models for each case study- a model for the three political candidates, a second model for the three candidates, and a third model for the three epidemics.

Next, we used statistical indicators of fit to assess the optimal number of topics for each case study. For each corpus, the following process was carried out. First, a range of models containing 10–300 topics were estimated. Due to the time-consuming nature of this process, models were examined by “skips” of 10 for models between 10–100, “skips” of 20 for models between 120–200, and two models containing 250 and 300 topics respectively. We then used this output to choose a smaller range of k-topics to focus on, based on four different model fit indicators. We chose the k-areas where indicators of accuracy (Arun, Suresh, Veni Madhavan, & Narasimha Murthy, 2010), and density (Cao, Xia, Li, Zhang, & Tang, 2009) were minimized, and indicators from latent concept modeling (Deveaud, Sanjuan, & Bellot, 2014) and Bayesian Markov chain Monte-Carlo algorithm (Griffiths & Steyvers, 2004) we maximized. For the chosen limited range, we reran the models without skips (e.g., after the first iteration pointed to the range of 30–80 for Senate candidates dataset, we modeled each of the possible 51 models for this corpus; 30, 31, 32, 33, etc.). Finally, we chose the final number of topics from that output based on the indicators detailed above. In terms of model resolution, we suggest researchers employing ANTMN will prefer to err on the side of too-many-topics rather than too few. This is as duplicate topics resulting from too high of a resolution can be grouped together in the community detection stage. However, resolution which is too low may increase the likelihood of hybrid topics (two distinct topics connected together), which cannot be separated to differing communities.

To interpret the topics, we examined three types of information; the words with the highest loading over each topic, the words that are both prevalent and exclusive to each topic (see definition and formula of FREX words in Roberts et al., 2014), and the full documents that are most representative of each topic. Two independent coders analyzed and labeled each topic based on a close reading of the word lists and the full documents. As is often the case in topic models, not all topics were relevant or interpretable, for several reasons. First, some topics resulted from articles that used the keywords in an irrelevant way (e.g., a real-estate article mentioning ‘Swiss Drive’ is not about the foreign country Switzerland). Second, “boilerplate” topics that have no substantive meaning and are linguistic artifacts were removed (DiMaggio et al., 2013; Maier et al., 2018). An example is a topic consisting of the top words us, day, time, one, years, just, know, back, people, and never in the political candidates corpus). Appendix 2 includes the top exclusive words for each topic for all three models constructed in the three case studies to allow the readers to get a better understanding of the various topics used in the networks, beyond the title assigned to them by the researchers.

Network Parameters

To create the network, we calculated pairwise cosine similarity between topics based on the documents they share. As mentioned earlier, each document is a mixture of topics. The information regarding this mixture is described by the theta matrix, in which columns represent the various topics and the rows represent the documents (Maier et al., 2018). Based on the clustering logic of framing analysis described in the theory section (van Gorp, 2010), and our conceptualization of topics as frame elements, we used the theta matrix to calculate the distance/similarity between each pair of topics – estimating the extent to which topics tend to appear in the same documents.

While outside the scope of this paper, there are other ways one could calculate topics similarity, for example by the extent to which topics share similar vocabularies (similarity over the beta matrix). In our experimentation with the three case studies we found that while minor differences do exist between the two methods, their resulting communities remain roughly the same. This similarity is likely due to the relationship between the theta and beta matrices, with the word loading over each topic in the beta matrix contributing to the calculation of the topic loadings for each document in the theta matrix. In addition to choosing between the theta and beta matrices, researchers could also use other measures of relationship between topics instead of cosine similarity, such as Pearson correlations. We found the results of cosine similarity and Pearson correlation to be highly similar

($r = 0.9$). However, while cosine-similarity ranges for positive values is positive (0–1), correlation values range between -1 and 1 . Due to most network community detection algorithm being unable to handle negative values we chose to use the cosine similarity measurement. However, further research is required in order to assess the viability of the two approaches. Researchers choosing to use Pearson correlation in the future are advised to pay attention to appropriate thresholding, or transformation techniques, required to incorporate negative edge values in the resulting network.

Next, based on the theta-cosine matrix, we constructed a network in which topics served as nodes and similarities between them as edges. This procedure offers both an efficient graphic representation of the estimated models (Maiya & Rolfe, 2014; Zhang, Li, & Zhang, 2018) and a rich data source for further analyses (on the node, edge, community, and network levels). We elaborate on both advantages at the discussion section. The result is a fully connected (all topics have a positive relationship), undirected (a symmetric relationship is assumed between pairs of topics in this bag of words approach), weighted network, where edges provide information on the extent to which topics are related (between 0 and 1).

Community Detection

A variety of algorithms for analyzing community structure of networks are available to researchers (Lancichinetti & Fortunato, 2009), all of which employ a division of a network's nodes into clusters that maximize within-group links, while minimizing between-group links. In our case, topics were clustered based on their prevalence in similar documents (as expressed in the theta matrix). We estimated how sensitive the community structure is to the use of 5 different detection algorithms and present the results in Appendix 1 for transparency. This sensitivity analysis demonstrated that various community detection algorithms produced virtually similar results, differing mostly by subdivisions of large communities into smaller components in some methods due to resolution differences. For example, a single community detected using a particular method might break down into two sub-components when using another algorithm.

The choice of community detection algorithm is a complicated one with various algorithms employing different approaches for identifying community structures in networks. This choice should also be based on network characteristics, which in the case of our topic-networks are: weighted, undirected, dense, and relatively small network (even for large corpora, number of topics tends to be small). Drawing on recent comparative analyses (Yang et al., 2016) we decided to employ the hierarchical clustering algorithm suggested by Pons and Latapy (2005; often referred to as “Walktrap”), the multi-level modularity maximization method suggested by Blondel et al. (2008; often referred to as “Louvain”), the Spinglass algorithm suggested by (Reichardt & Bornholdt, 2006), and a Fast-Greedy algorithm proposed by Clauset, Newman, and Moore (2004). Importantly, as the choice of algorithms should also be guided by the research question and the ability of the method to provide interpretable results (Fortunato & Hric, 2016; Yang et al., 2016) we decided to add the Leading Eigenvector algorithm suggested by Newman (2006), as we found it to be an effective tool for analyzing our three corpora. For parsimony, we focus in the results section on the popular Walktrap algorithm (Pons & Latapy, 2005), due to its computational efficiency and tendency to offer a smaller number of communities, thus being an efficient tool for data reduction purposes. However, we urge researchers employing our method to utilize other suitable algorithms for sensitivity analysis. We elaborate more on this issue and present our preliminary conclusions from the three case studies analyzed here in the discussion section.

Results

Senate Candidates

Our topic model for the three Republican candidates consisted of 66 topics. The Walktrap clustering resulted in two communities. As can be seen in Figure 1, the first community on the left focuses on voting, polls, the presidential race, the media, debates and campaigns. Representative articles from these topics portrayed, for example, worries among the Republican candidates about Trump's nomination

topic of the New Hampshire race that was assigned to the right community in Walktrap moved to the left one in Spinglass, and the climate change topic moved from the right to the left. Most topics remained in the same cluster.

Foreign Countries

The topic model for the three foreign countries (Egypt, Brazil, and Switzerland) corpus consisted of 76 topics. As can be seen in Figure 2, a Walktrap community detection algorithm identified five communities. Topics in the first community (bottom right) were about issues of domestic and international conflict, often violent ones. This large community included the topics of the Iranian nuclear program, conflicts between Israel and Palestine, the Arab Spring, and US policy interventions in the Ukrainian war. A neighbor community, (top right corner) consisted of topics about diplomatic relations, international courts and trials (e.g., related to Guantanamo or to American journalists accused by Iranian courts of being spies), and the police. The third community (middle) contained a group of topics relating to economic issues and sports. The top side of this cluster focuses on economy, for example the 2008 recession, stocks, financial reports, energy and oil, budgets, and governmental spending. The bottom of this community is about sports, for example about soccer or tennis. It is important to note that while the visualization of the network is useful for analysis it does have some limitation due to the multidimensional nature of networks being presented in a 2-dimensional space. Thus, visual distance in this visualization of the networks (as the ones in Figure 1–3) is an approximation of topic distance rather than an exact one. The fourth community (top left) contained a collection of topics with a unifying theme of culture. These included various cultural aspects such as film and book reviews, home design and fashion, food, tourism, and science. Lastly, the small community on the left addresses other cultural

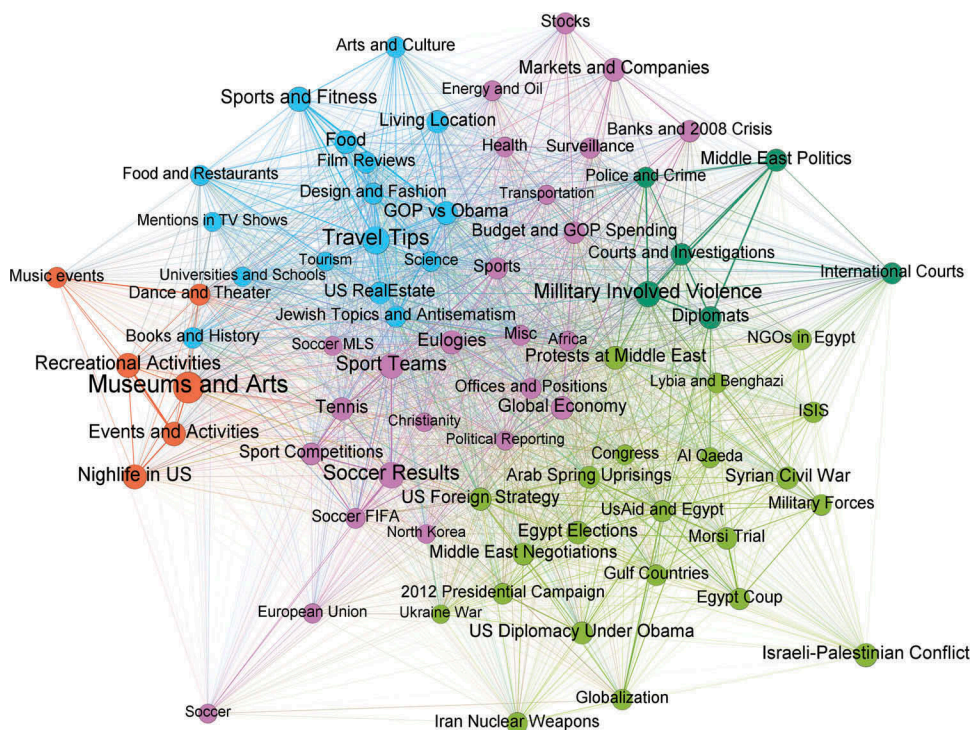


Figure 2. The topic network drawn from the coverage of the three countries (Brazil, Egypt, Switzerland) between the years 2010–2015. Nodes represent topics; Edges represent co-occurrence of topics in documents (calculated using cosine similarity); Size represents the prominence of each topic; Color represents community membership. Network is weighted, fully connected and undirected.

aspects, with a focus on specific recreational events, such as musical shows and theater plays, and interactions between the local and foreign countries (e.g., recreational events that borrow on foreign countries' culture, such as Egyptian beads and Brazilian Capoeira classes). As can be seen in the Appendix 1, the network remained relatively stable with different community detection algorithms, with changes tending to occur in the border between the right community and the left ones (i.e., mostly in the community of economy and sports).

Epidemics

The epidemics model consisted of 53 topics. The Walktrap community detection resulted in three communities. As can be seen in Figure 3, the largest community on the right consisted of topics about the cultural and economic consequences of diseases and about political conflicts. Articles dominated by these topics portrayed the impact of diseases on the economy at large (e.g., stocks and markets), on specific industries (tourism) and cultural events (Rio Olympic Games), and on politicians (Mayor De Blasio). Solutions were also discussed in these articles, such as the allocation of resources (funding), voting on laws (Senate and congress), impeaching those who failed to adequately cope with the diseases (Rousseff's impeachment), nominating response teams (Obama), sending support abroad (military aid for Ebola), distributing vaccines (H1N1 vaccine distribution), and conducting research (scientific breakthroughs). The community on the left dealt with diseases abroad that threatened to invade the United States, and how health organizations such as CDC, are

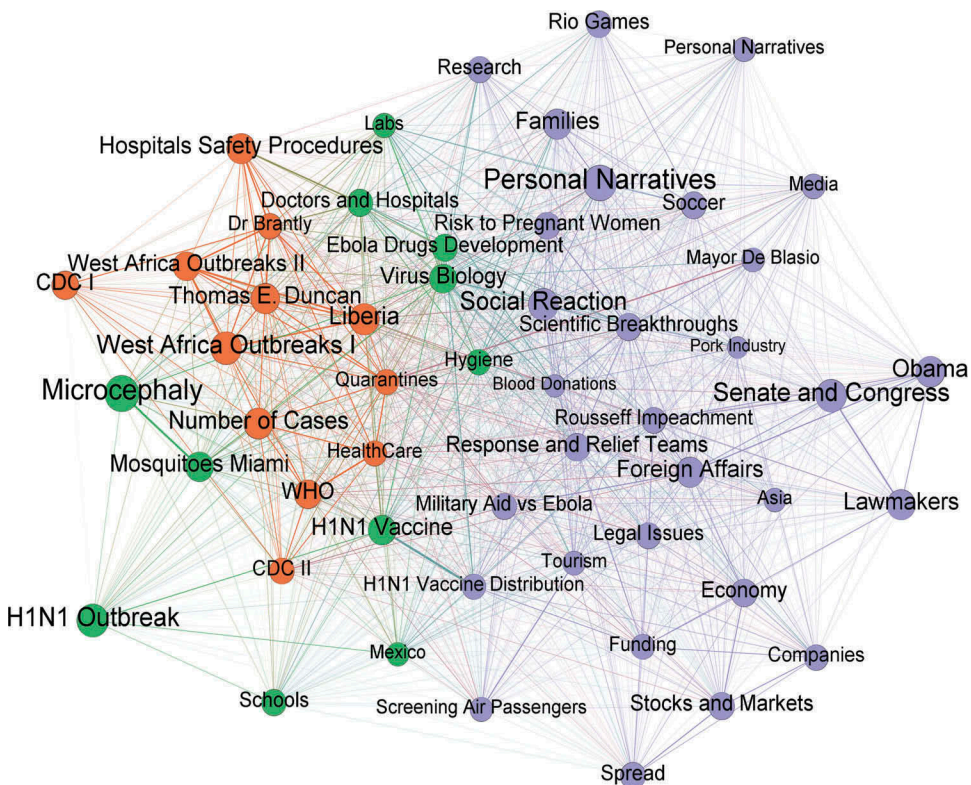


Figure 3. The topic network drawn from the coverage of the three epidemics (H1N1 of 2009, Ebola of 2014, and Zika of 2016). Nodes represent topics; Edges represent co-occurrence of topics in documents (calculated using cosine similarity); Size represents the prominence of each topic; Color represents community membership. Network is weighted, fully connected and undirected.

trying to stop the spread into the States. It included topics about specific infected individuals, such as Thomas E. Duncan, who brought Ebola to the U.S, and topics about the doctors who treated them. Some topics are about safety procedures in hospitals, such as quarantines. The community in the middle of the network included topics about biological facts, including the health effects of diseases (e.g., microcephaly for Zika), symptoms, and the spread of diseases (e.g., mosquitoes). The network remained relatively the same with different algorithms, though some methods, such as Louvain, broke the left community to sub-communities (See Appendix 1).

Discussion

Framing theory is one of the most central, applicable, and contested theories in communication studies (Entman, 1993). In recent years, researchers have been debating the conceptualization and operationalization of framing in attempts to increase validity and reliability (Cacciatore et al., 2016; Matthes & Kohring, 2008). Among the main worries expressed was the question of what constitutes a frame and how frames could be measured with minimal bias. One suggested solution was to use inductive analysis of frame components, and to harness computational tools that could automatically cluster components into frame packages based on linguistic patterns (Matthes & Kohring, 2008; van Gorp, 2010).

With the introduction of ANTMN method, we suggest that frames could be inductively identified and analyzed using a three-step process, consisting of topic modeling, network analysis, and community detection. Our method can be used in cases where a dominant paradigm for frame analysis exists, in ones where researchers have gathered evidence for linguistic patterns in coverage without defining them as frames, and where knowledge about coverage is relatively scarce. To examine its applicability, we employed ANTMN to the analysis of three case studies, representing each condition. The first analysis of political candidates during campaigns, relied on a highly cited approach, namely the use of strategy and issue framing (Cappella & Jamieson, 1997). A second analysis looked into the coverage of soft and hard power aspects in the coverage of foreign nations, that while prevalent in the literature, has only sporadically been considered as constituting news frames (Chitty, 2017; Nye, 2004). The third analysis examined an area where little evidence is available regarding consistent coverage patterns across case studies, that is, the coverage of epidemics (the authors, 2018b). In all three case studies, our method identified distinct and coherent topic communities that are consistent with the linguistic conceptualization of framing, while also being intelligible for the researchers to interpret.

For the political candidates corpus, we found the two communities detected by ANTMN to be highly consistent with the common conceptualization of strategy and issue frames (Cappella & Jamieson, 1997). The strategy community on the left of Figure 1 was composed of the components theorized previously as the building blocks of strategy framing (Aalberg, Strömbäck, & De Vreese, 2011; Cappella & Jamieson, 1997); competitions, mention of performers and audiences (voters), individual performance, personal attacks and individual characteristics (e.g., criticism of Trump's character), as well as polls and position in the "horse race". The second community represented discourses about specific policy (issue framing), such as climate and energy, education, gun control, economy, immigration, healthcare, and foreign affairs (Aalberg et al., 2011; Zoizner, 2018).

In the foreign nations case study, our method identified five communities. Though a framing approach to foreign countries coverage is lacking, the literature does point to prominent patterns in the coverage of foreign countries (Kaneva, 2011), most notably to the focus on hard and soft power (Hayden, 2012; Nye, 2004). Hard power is employed through military force and coercion, while soft power refers to the capacity to wield influence on a target audience to elicit desirable outcomes by means of culture, art, music, performance in sports, or economy (Nye, 2011). In our results, the largest community on the bottom left side of Figure 2 and the one next to it on the top right seem consistent with the definition of hard power, with a focus on conflicts and diplomatic negotiations, military interventions and coercive diplomacy (Wilson, 2008). The two communities on the left seem congruent with soft power, discussing cultural aspects, such as tourism and cultural heritage (Anholt, 2009; Fan, 2010). Consistent with prior theorizations of economy being both a matter of soft and hard power

(Dinnie, 2009), the economic community in our model was situated in between the other communities. Interestingly, sports topics were clustered together with the economy. We postulate it might be due to the unique use of sports and game language when discussing both soft and hard power (e.g., a country could “defeat” ISIS, and a soccer team could “defeat” its opponent).

As explained earlier, the literature about media coverage of epidemics lacks a consensus among researchers regarding coverage patterns, with different researchers focusing on different aspects of coverage (The authors, 2018b). We, therefore, attempted to understand the results of our network through multiple theoretical perspectives suggested by different scholars and not based on one dominant perspective. The epidemics network consisted of three communities. The first included topics about the political, economic, cultural, and social aspects of the disease. This community is consistent with the economic frame (Tian & Stewart, 2005; Ophir, 2018b) and the conflict frame found in some previous studies (An & Gower, 2009). The frame elements included, for example, topics about Zika’s effect on the Rio Olympics and the Brazilian government (two separate topics), or H1N1’s impact on the pork industry. The second community was consistent with the medical frame (Tian & Stewart, 2005; the authors, 2018b) and included frame elements such as statistics (e.g., death tolls), symptoms and health effects, or transmission routes. Lastly, the third community discussed diseases abroad that threatened to invade the United States and officials’ attempts to block it from happening. This pattern was discussed in our previous work (the authors, 2018b), but did not gain much attention in the literature in general.

As for the question of whether or not what ANTMN identified is media frames or other textual phenomenon, we find that the methodological steps ANTMN follows are consistent with the conceptualization of frames as news patterns that “repeatedly invoke the same objects and traits, using identical or synonymous words and symbols in a series of similar communications that are concentrated in time” (Entman et al., 2009, p. 177). We do find support for the validity of our method in the fact that it yielded virtually identical results to what scholars have consistently found using other approaches in the extensively-validated area of political campaigns and the use of strategy and issue. More research using other validated framing theories will bolster our confidence even further. It should be noted that just as the conceptual meaning of topics is sensitive to their number (k), so will the meaning of network communities of topics depend, in part, on the number of clusters and the community detection method chosen. However, in our empirical testing of the communities resulting from different methods, we found that in opposition to topics, the communities tended to be stable and fairly consistent across methods. In many cases, communities were almost identical between methods, and in other cases, the differences were relatively small (e.g., a difference between three and four communities). We believe that current conceptualizations of frames cannot tell researchers exactly how wide or narrow frames should be, and unfortunately, no method could do it either. Instead, the choice of an optimal level of abstraction (resolution) is a theoretical question that should be debated by scholars and experts at the specific context where the method is used, based on research needs. We do hope that the use of ANTMN by future researchers will contribute to our understanding of these questions about news frames.

Interestingly, our method identified context-specific frames and not generic ones, though the distinction between the two types remains vague (D’Angelo, 2018) and our findings may result from the specific case studies we selected. Specifically, whether or not the frames identified in this demonstration could be used and found in other journalistic context is an empirical question that could only be answered by using the method in more studies on more contexts.

To summarize, in the first case-study, exploring the news coverage of political candidates, our method resulted in frames that were very consistent with issue and strategy framing, though with greatly reduced resources compared to methods such as manual content analysis. When examining the news coverage of foreign countries, it also supported the yet-to-be-established argument that soft and hard power could be seen as news frames. However, it also offered a more detailed and high-resolution view of these two broad categories, highlighting different critical facets of soft and hard

power. Finally, when examining the news coverage of epidemics, all frames identified were novel and were not previously discussed in literature.

These results demonstrate the ability of the ANTMN method to identify and measure the prevalence of frame elements (in the form of topics, though in some cases several elements appear in the same topic) and frame packages (topic communities). Though we suggest researchers will interpret the results of their findings relying on prior knowledge about the media and the area of coverage, as we did here, ANTMN's way of building the topic network is purely inductive and does not rely on any prior knowledge. Instead, the number of topics in each model was selected based on statistical indicators, and clustering of elements together was done automatically using community detection algorithms (relying on default hyper-parameters stated in the most common R applications of these algorithms). As is always the case in unsupervised machine learning, the last stage of analysis, the hermeneutic process of interpreting the findings, remains the work of expert humans who are familiar with the specific field (DiMaggio et al., 2013). Nevertheless, our method lays out a map of the corpus for the researcher to analyze, thus making one more crucial step toward reliance on textual evidence, and the reduction of bias and subjectivity in framing analysis (van Gorp, 2010).

Our proposed method has several advantages over existing techniques used for identifying and measuring frames. First, compared with manual identification of frames, our method requires considerably less resources in terms of the money and time associated with manual coding. Second, it reduces researcher biases. Third, it allows for an inductive exploration of frame structures and the discovery of frames that were not detected in previous studies and therefore are in risk of being overlooked. Fourth, it builds upon advancements made in prior attempts to use computational tools and unsupervised machine learning approaches to frame-analysis (Baden, 2010, 2018; Diesner & Carley, 2011; DiMaggio et al., 2013; Doerfel & Barnett, 1999), but makes further progress by allowing for multiple membership of words in various topics and avoiding treating individual topics as frames.

Perhaps the greatest advantage of this approach is that it opens a door for future adoption of more indicators and techniques used in the rapidly developing field of network analysis. We focus our analysis on the single technique of community detection. However, recent developments in network analysis included more potential tools for communication researchers to adopt for the analysis of the relationships between topics. For example, one could analyze the centrality of topics or the structural role they play as important junctions in the network through which information passes (Rubinov & Sporns, 2010). Several additional tools are currently being developed by various scholars, including techniques of multiple membership community detection, and multi-level networks (Fortunato & Hric, 2016). These tools will allow for even more nuanced analysis of discourse, for example of the dynamic development of frame structures over time or across contexts.

Several caveats should be taken into consideration. First, while our method could automatically identify frame elements and packages, the interpretation of findings still requires a qualitative analysis by experts in the field of interest. Second, our method may be limited to the analysis of emphasis frames and might not be suitable for analyzing equivalency frames. Third, as demonstrated in each of our case studies, textual corpora are bound to have irrelevant topics due to the nature of stylistic conventions and the algorithm itself. The step of deciding which topics to omit from the network inevitably affects the final results and should be done carefully and conservatively. However, these artifacts existed in all prior attempts to analyze frames, and were mostly ignored, especially by those conducting manual content analysis. Fourth, as is the case with many machine learning techniques, topic modeling and some community detection techniques are probabilistic, meaning that results may vary somewhat between iterations. While this could be ameliorated by averaging the results of several iterations, this procedure is likely to be very taxing in terms of time and computational resources, especially for large corpora. In addition, in our own experience, the networks presented here are very stable across iterations. Lastly, we have limited the scope of our corpora as our aim here was to introduce and demonstrate a method and not to study the coverage of specific topics such as epidemics. A thorough analysis of these substantive topics might require different

corpora with a larger sample of candidates, countries or epidemics, better external validity or theoretical frameworks beyond framing.

The approach taken here paves the way for future inquiries. For example, we built the matrices based on the use of different topics in similar documents. Other approaches may correlate topics by other means, for example by their vocabulary similarity (that is over the beta matrix instead of the theta matrix used here). Similarly, in this study we compared the results obtained from multiple community detection algorithms to examine how sensitive the results are to different methods. For that purpose, we used algorithms that were recommended in recent reviews for small, undirected, weighted networks: Walktrap, Louvain, Spinglass, and Fast-Greedy. In addition, we found the Leading Eigenvector algorithm suggested by Newman (2006) to be effective in creating intelligible clusters, that tended to be similar to Walktrap's output. The results, as can be seen in Appendix 1, indicate that, generally speaking, community structures were stable across methods. That was especially the case with the political candidates' case study, where minimal differences were found between solutions of the five algorithms. In the second and third case studies, most algorithms produced similar communities, with some algorithms resulting in more communities (e.g., Louvain and Fast-Greedy) than others (e.g., Walktrap and Eigenvector). As differences were not dramatic, and due to its consistently parsimonious outputs, we recommend the use of Walktrap for framing analysis based on our examination. However, as in the case of relationship matrices (theta vs. beta), and topic modeling techniques (e.g., use the alternative of Structural Topic Modeling, Roberts et al., 2014), future studies using more case studies may be able to provide a systematic description of the differences between algorithms.

To summarize, we introduce and demonstrate the efficiency and applicability of a new method for the inductive identification and measurement of media frames in large and complex corpora using unsupervised machine learning techniques. We suggest that the ANTMN method could reduce bias and costs that were associated with prior methods for identification and measurement of frames. We demonstrate the importance of connecting novel computational tools to established theoretical perspectives, a symbiosis that could greatly improve our understanding of social phenomena.

Note

1. It should be noted that in our own previous work we have used other methods for topic modeling such as Structural topic modeling (Roberts et al., 2014) and found similar results. However, further research is needed to determine the applicability of our method to alternative algorithms.

Disclosure statement

No potential conflict of interest was reported by the author.

ORCID

Dror Walter  <http://orcid.org/0000-0001-9870-998X>
Yotam Ophir  <http://orcid.org/0000-0001-5049-7906>

References

- Aalberg, T., Strömbäck, J., & De Vreese, C. H. (2011). The framing of politics as strategy and game: A review of concepts, operationalizations and key findings. *Journalism*, 13, 162–178. doi:10.1177/1464884911427799
- An, S.-K., & Gower, K. K. (2009). How do the news media frame crises? A content analysis of crisis news coverage. *Public Relations Review*, 35, 107–112. doi:10.1016/j.pubrev.2009.01.010
- Anholt, S. (2009). The media and national image. *Place Branding and Public Diplomacy*, 5, 169–179. doi:10.1057/pb.2009.11
- Arun, R., Suresh, V., Veni Madhavan, C. E., & Narasimha Murthy, M. N. (2010). On finding the natural number of topics with latent dirichlet allocation: Some observations. In M. J. Zaki, J. X. Yu, B. Ravindran, & V. Pudi (Eds.),

- Proceedings of the 14th Pacific-Asia conference on advances in knowledge discovery and data mining Part I*, (pp. 391–402). Berlin, Germany: Springer.
- Baden, C. (2010). *Communication, contextualization & cognition: patterns & processes of frames' influence on people's interpretations of the EU constitution* (Unpublished Doctoral Dissertation). University of Amsterdam, Amsterdam, the Netherlands.
- Baden, C. (2018). Reconstructing frames from intertextual news discourse: A semantic network approach to news framing analysis. In P. D'Angelo (Ed.), *Doing news framing analysis II: Empirical and theoretical perspectives* (pp. 43–66). New York, NY: Routledge.
- Blei, D. M. (2012). Probabilistic topic models. *Communications of the ACM*, 55, 77–84. doi:10.1145/2133806.2133826
- Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *The Journal of Machine Learning Research*, 3, 993–1022.
- Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 10, 1–12. doi:10.1088/1742-5468/2008/10/P10008
- Boydston, A. E., & Glazier, R. A. (2013). A two-tiered method for identifying trends in media framing of policy issues: The case of the war on terror. *Policy Studies Journal*, 41, 707–736. doi:10.1111/psj.12038
- Cacciatore, M. A., Scheufele, D. A., & Iyengar, S. (2016). The end of framing as we know it ... and the future of media effects. *Mass Communication and Society*, 19, 7–23. doi:10.1080/15205436.2015.1068811
- Cao, J., Xia, T., Li, J., Zhang, Y., & Tang, S. (2009). A density-based method for adaptive LDA model selection. *Neurocomputing*, 72, 1775–1781. doi:10.1016/j.neucom.2008.06.011
- Cappella, J. N., & Jamieson, K. H. (1997). *Spiral of cynicism: The press and the public good*. New York, NY: Oxford University Press.
- Chitty, N. (2017). *The Routledge handbook of soft power*. New York, NY: Routledge.
- Chong, D., & Druckman, J. N. (2007). A theory of framing and opinion formation in competitive elite environments. *Journal of Communication*, 57, 99–118. doi:10.1111/j.1460-2466.2006.00331.x
- Clauset, A., Newman, M. E. J., & Moore, C. (2004). Finding community structure in very large networks. *Physical Review E*, 70, 066111-1 – 06111-6. doi:10.1103/PhysRevE.70.066111
- Cohen, B. C. (1965). *The press and foreign policy*. Princeton, NJ: Princeton University Press.
- Curtis, B. (2012). Understanding tailored internet smoking cessation messages a reasoned action approach. *The Annals of the American Academy of Political and Social Science*, 640, 136–149. doi:10.1177/0002716211423642
- D'Angelo, P. (2018). *Doing news framing analysis II: Empirical and theoretical perspectives*. New York, NY: Routledge.
- De Vreese, C. H. (2005). The spiral of cynicism reconsidered. *European Journal of Communication*, 20, 283–301. doi:10.1177/0267323105055259
- De Vreese, C. H. (2010). Framing the economy: Effects of journalistic news frames. In P. D'Angelo & J. A. Kuypers (Eds.), *Doing news framing analysis: Empirical and theoretical perspectives* (pp. 203–230). New York, NY: Routledge.
- Deveaud, R., Sanjuan, E., & Bellot, P. (2014). Accurate and effective latent concept modeling for ad hoc information retrieval. *Revue Des Sciences Et Technologies De l'Information - Série Document Numérique*, 61–84. doi:10.3166/DN.17.1.61-84
- Diesner, J., & Carley, K. M. (2011). Semantic networks. In G. Barnett (Ed.), *Encyclopedia of social networks, Volume 2* (pp. 766–769). Thousand Oaks, CA: Sage Publication.
- DiMaggio, P., Nag, M., & Blei, D. (2013). Exploiting affinities between topic modeling and the sociological perspective on culture: Application to newspaper coverage of U.S. government arts funding. *Poetics*, 41, 570–606. doi:10.1016/j.poetic.2013.08.004
- Dinnie, K. (2009). *Nation branding: Concepts, issues, practice*. Amsterdam, the Netherlands: Butterworth-Heinemann.
- Doerfel, M. L., & Barnett, G. A. (1999). A semantic network analysis of the international communication association. *Human Communication Research*, 25, 589–603. doi:10.1111/j.1468-2958.1999.tb00463
- Entman, R. M. (1993). Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, 43, 51–58. doi:10.1111/j.1460-2466.1993.tb01304.x
- Entman, R. M., Matthes, J., & Pellicano, L. (2009). Nature, sources, and effects of news framing. In K. Wahl-Jorgensen & T. Hanitzsch (Eds.), *The handbook of journalism studies* (pp. 175–190). New York, NY: Routledge.
- Fan, Y. (2010). Branding the nation: Towards a better understanding. *Place Branding and Public Diplomacy*, 6, 97–103. doi:10.1057/pb.2010.16
- Fortunato, S., & Hric, D. (2016). Community detection in networks: A user guide. *Physics Reports*, 659, 1–44. 1608.00163. doi:10.1016/j.physrep.2016.09.002.
- Galtung, J., & Ruge, M. H. (1965). The structure of foreign news: The presentation of the Congo, Cuba and Cyprus crises in four Norwegian newspapers. *Journal of Peace Research*, 2, 64–90. doi:10.1177/002234336500200104
- Gamson, W. A., & Modigliani, A. (1989). Media discourse and public opinion on nuclear power: A constructionist approach. *American Journal of Sociology*, 95, 1–37. doi:10.2307/2780405
- Gitlin, T. (1980). *The whole world is watching: Mass media in the making & unmaking of the new left*. Berkeley: University of California Press.
- Griffiths, T. L., & Steyvers, M. (2004). Finding scientific topics. *Proceedings of the National Academy of Sciences*, 101, 5228–5235. doi:10.1073/pnas.0307752101

- Grimmer, J., & Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21, 267–297. doi:10.1093/pan/mps028
- Hayden, C. (2012). *The rhetoric of soft power: Public diplomacy in global contexts*. Plymouth, United Kingdom: Lexington Books.
- Inglehart, R., & Welzel, C. (2010). Changing mass priorities: The link between modernization and democracy. *Perspectives on Politics*, 8, 551–567. doi:10.1017/S1537592710001258
- Iyengar, S. (1996). Framing responsibility for political issues. *The Annals of the American Academy of Political and Social Science*, 546, 59–70. doi:10.1177/0002716296546001006
- Ji, L. (2017). Measuring soft power. In N. Chitty, L. Ji, G. D. Rawnsley, & C. Hayden (Eds.), *The routledge handbook of soft power* (pp. 75–93). New York, NY: Routledge.
- Kahneman, D., & Tversky, A. (1984). Choices, values, and frames. *American Psychologist*, 39, 341. doi:10.1037/0003-066X.39.4.341
- Kaneva, N. (2011). Nation branding: Toward an agenda for critical research. *International Journal of Communication*, 5, 25.
- Klebanov, B. B., Diermeier, D., & Beigman, E. (2008). Automatic annotation of semantic fields for political science research. *Journal of Information Technology & Politics*, 5, 95–120. doi:10.1080/19331680802149640
- Lancichinetti, A., & Fortunato, S. (2009). Community detection algorithms: A comparative analysis. *Physical Review E*, 80. doi:10.1103/PhysRevE.80.056117
- Maier, D., Waldherr, A., Miltner, P., Wiedemann, G., Niekler, A., Keinert, A., ... Adam, S. (2018). Applying LDA topic modeling in communication research: Toward a valid and reliable methodology. *Communication Methods and Measures*, 12, 93–118. doi:10.1080/19312458.2018.1430754
- Maiya, A. S., & Rolfe, R. M. (2014). Topic similarity networks: Visual analytics for large document sets. Paper presented at IEEE International Conference on Big Data, 2014. Washington, DC, IEEEExplore Digital Library.
- Matthes, J. (2009). What's in a frame? A content analysis of media framing studies in the world's leading communication journals, 1990–2005. *Journalism & Mass Communication Quarterly*, 86, 349–367. doi:10.1177/107769900908600206
- Matthes, J., & Kohring, M. (2008). The content analysis of media frames: Toward improving reliability and validity. *Journal of Communication*, 58, 258–279. doi:10.1111/j.1460-2466.2008.00384.x
- McCombs, M. E., & Shaw, D. L. (1972). The agenda-setting function of mass media. *Public Opinion Quarterly*, 36, 176–187. doi:10.1086/267990
- Mimno, D., & Blei, D. (2011). Bayesian checking for topic models. *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, Edinburgh, Scotland, UK, 227–237.
- Nelson, T. E., Clawson, R. A., & Oxley, Z. M. (1997). Media framing of a civil liberties conflict and its effect on tolerance. *The American Political Science Review*, 91, 567. doi:10.2307/2952075
- Newman, M. E. J. (2006). Finding community structure in networks using the eigenvectors of matrices. *Physical Review E*, 74. doi:10.1103/PhysRevE.74.036104
- Nisbet, M. C. (2010). Knowledge into action: Framing the debates over climate change and poverty. In P. D'Angelo & J. A. Kuypers (Eds.), *Doing news framing analysis: Empirical and theoretical perspectives* (pp. 43–83). New York, NY: Routledge.
- Nye, J. S., Jr. (2004). *Soft power: The means to success in world politics*. New York, NY: Public Affairs.
- Nye, J. S., Jr. (2011). *The future of power*. New York, NY: Perseus.
- Ophir, Y. (2018a). Spreading news: The coverage of epidemics by American newspapers and its effects on audiences - A crisis communication approach (Doctoral Dissertation). University of Pennsylvania, Philadelphia, US.
- Ophir, Y. (2018b). Coverage of epidemics in american newspapers through the lens of the crisis and emergency risk communication framework. *Health Security*, 16, 147–157. doi: 10.1089/hs.2017.0106
- Pons, P., & Latapy, M. (2005). Computing communities in large networks using random walks. In P. Yolum, T. Güngör, F. Gürgen, & C. Özturan (Eds.), *Computer and information sciences - ISCIS 2005* (pp. 284–293). Berlin, Germany: Springer.
- Puschmann, C., & Scheffler, T. (2016). Topic modeling for media and communication research: A short primer. SSRN Scholarly Paper ID 2836478. Rochester, NY Social Science Research Network. Retrieved from: <https://papers.ssrn.com/abstract=2836478>. 10.2139/ssrn.2836478
- Reichardt, J., & Bornholdt, S. (2006). Statistical mechanics of community detection. *Physical Review E*, 74. doi:10.1103/PhysRevE.74.016110
- Reynolds, B., & Seeger, M. W. (2014). Crisis and emergency risk communication, 2014 edition. Centers for Disease Control and Prevention. Retrieved from https://emergency.cdc.gov/cerc/resources/pdf/cerc_2014edition.pdf
- Roberts, M., Stewart, B. M., Tingley, D., Lucas, C., Leder-Luis, J., Gadarian, S. K., ... Rand, D. G. (2014). Structural topic models for open-ended survey responses. *American Journal of Political Science*, 58, 1064–1082. doi:10.1111/ajps.12103
- Rubinov, M., & Sporns, O. (2010). Complex network measures of brain connectivity: Uses and interpretations. *NeuroImage*, 52, 1059–1069. doi:10.1016/j.neuroimage.2009.10.003

- Schofield, A., Thompson, L., & Mimno, D. (2017). Quantifying the effects of text duplication on semantic models. *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, Copenhagen, Denmark, 2737–2747. doi:[10.18653/v1/D17-1290](https://doi.org/10.18653/v1/D17-1290)
- Semetko, H. A., & Valkenburg, P. M. (2000). Framing European politics: A content analysis of press and television news. *Journal of Communication*, 50, 93–109. doi:[10.1111/j.1460-2466.2000.tb02843.x](https://doi.org/10.1111/j.1460-2466.2000.tb02843.x)
- Shah, D. V., McLeod, D. M., Gotlieb, M. R., & Lee, N.-J. (2009). Framing and agenda setting. In R. L. Nabi & M. B. Oliver (Eds.), *The SAGE handbook of media processes and effects* (pp. 83–98). Thousand Oaks, CA: Sage Publication.
- Shih, T., Wijaya, R., & Brossard, D. (2008). Media coverage of public health epidemics: Linking framing and issue attention cycle toward an integrated theory of print news coverage of epidemics. *Mass Communication and Society*, 11, 141–160. doi:[10.1080/15205430701668121](https://doi.org/10.1080/15205430701668121)
- Tankard, J. W. J. (2001). The empirical approach to the study of media framing. In S. D. Reese, O. H. Gandy Jr., & A. E. Grant (Eds.), *Framing public life: Perspectives on media and our understanding of the social world* (pp. 95–106). Mahwah, NJ: Lawrence Erlbaum Associates.
- Tian, Y., & Stewart, C. M. (2005). Framing the SARS crisis: A computer-assisted text analysis of CNN and BBC online news reports of SARS. *Asian Journal of Communication*, 15, 289–301. doi:[10.1080/01292980500261605](https://doi.org/10.1080/01292980500261605)
- Tuchman, G. (1973). Making news by doing work: Routinizing the unexpected. *American Journal of Sociology*, 79, 110–131. doi:[10.1086/225510](https://doi.org/10.1086/225510)
- van Atteveldt, W., & Peng, T.-Q. (2018). When communication meets computation: Opportunities, challenges, and pitfalls in computational communication science. *Communication Methods and Measures*, 12, 81–92. doi:[10.1080/19312458.2018.1458084](https://doi.org/10.1080/19312458.2018.1458084)
- van Atteveldt, W. H. (2008). *Semantic network analysis: Techniques for extracting, representing, and querying media content* (Unpublished Doctoral Dissertation). Amsterdam Vrije Universiteit, Amsterdam, the Netherlands.
- van Gorp, B. (2005). Where is the frame? Victims and intruders in the Belgian press coverage of the asylum issue. *European Journal of Communication*, 20, 484–507. doi:[10.1177/0267323105058253](https://doi.org/10.1177/0267323105058253)
- van Gorp, B. (2007). The constructionist approach to framing: Bringing culture back in. *Journal of Communication*, 57, 60–78. doi:[10.1111/j.1460-2466.2006.00329.x](https://doi.org/10.1111/j.1460-2466.2006.00329.x)
- van Gorp, B. (2010). Strategies to take subjectivity out of framing analysis. In P. D'Angelo & J. A. Kuypers (Eds.), *Doing news framing analysis: Empirical and theoretical perspectives* (pp. 100–125). New York, NY: Routledge.
- Wallach, H. M. (2006). Topic modeling: Beyond bag-of-words. *Proceedings of the 23rd International Conference on Machine Learning*, 977–984. doi: [10.1145/1143844.1143967](https://doi.org/10.1145/1143844.1143967)
- Walter, D. (2018). Exploring thematic diversity in news coverage and social media activity of political candidates using unsupervised machine learning (Doctoral Dissertation). University of Pennsylvania, Philadelphia, US.
- Walter, D., Sheaffer, T., Nir, L., & Shenhav, S. (2016). Not all countries are created equal: Foreign countries prevalence in U.S. news and entertainment media. *Mass Communication and Society*, 19, 522–541. doi:[10.1080/15205436.2016.1170853](https://doi.org/10.1080/15205436.2016.1170853)
- Wilson, E. J. (2008). Hard power, soft power, smart power. *The Annals of the American Academy of Political and Social Science*, 616, 110–124. doi:[10.1177/0002716207312618](https://doi.org/10.1177/0002716207312618)
- Wu, M. (2006). Framing AIDS in China: A comparative analysis of US and Chinese wire news coverage of HIV/AIDS in China. *Asian Journal of Communication*, 16, 251–272. doi:[10.1080/01292980600857781](https://doi.org/10.1080/01292980600857781)
- Yang, Z., Algesheimer, R., & Tessone, C. J. (2016). A comparative analysis of community detection algorithms on artificial networks. *Scientific Reports*, 6. doi:[10.1038/srep30750](https://doi.org/10.1038/srep30750)
- Zhang, C., Li, Z., & Zhang, J. (2018). A survey on visualization for scientific literature topics. *Journal of Visualization*, 21, 321–335. doi:[10.1007/s12650-017-0462-2](https://doi.org/10.1007/s12650-017-0462-2)
- Zoizner, A. (2018). The consequences of strategic news coverage for democracy: A meta-analysis. *Communication Research*, 1–23. doi:[10.1177/0093650218808691](https://doi.org/10.1177/0093650218808691)